# Zirong Gan

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Storrs, Connecticut- 06268, USA

## Research Interest

My research focuses on the intersection of AI and networking, spanning both AI for networks and networks for AI. I explore how machine learning and large language models can enable intelligent, automated networked systems, with an emphasis on network design, management, and optimization. I also investigate how network architectures and communication strategies can improve the performance, efficiency, and scalability of distributed AI systems.

## Experience

- **University of Connecticut [🌐]** *2025 - Present*
  *Research Assistant* Storrs, CT, USA
  - Developed P4Bench to evaluate the performance of LLMs in P4 programming.
  - Exploring the integration of LLMs with networking, focusing on Programmable Data Planes and P4 language.
- **Tencent Technology (Shenzhen) Co., Ltd. [🌐]** *2022 - 2025*
  *Backend Engineer* Shenzhen, China
  - Built Blockchain system, including consensus algorithm, smart contract.
  - Developed Web3.0 applications, including gaming, Blockchain explorer.
- **Jinninghui Technology(Blockchain Startup)** *2018 - 2022*
  *Blockchain Engineer* Guangzhou, China
  - Built the Blockchain system from the ground up.
  - Developed a developer platform for building applications on the Blockchain.

## Education

- **University of Connecticut** *2025 - Present*
  *Ph.D. School of Computing* Storrs, CT, USA
- **Sun Yat-sen University** *2014 - 2018*
  *B.E. School of Electronics and Information Technology* Guangzhou, China

## Publications

**[SIGMETRICS]** **Zirong Gan**, Shuyi Liao, Yannian Niu, Xinyue Zheng, Tingting Yu, Minmei Wang*. **P4Bench: A Scalable and Adaptive Oracle-Backed Benchmark for P4 Programming with Large Language Models**. In *Proceedings of the ACM SIGMETRICS International Conference on Measurement and Modeling of Computer Systems (SIGMETRICS '27)*.

## Skills

- **Programming Languages:** Golang, Python, C/C++, Java, Rust, Solidity, SQL, Bash
- **Database Systems:** MongoDB, PostgreSQL, MySQL, Redis, SQLite
- **DevOps & Version Control:** Docker, Git, Jenkins